

The economics of adversarial procurement in two-value space: intuition behind the knapsack reversal

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The author takes sole responsibility for all claims.

Abstract. This document develops the economic intuition behind the *knapsack reversal*: the finding that in adversarial procurement under a two-value structure, a simple value-maximisation rule (*value-first*, or ME) asymptotically outperforms the textbook cost-effectiveness heuristic (*ratio-greedy*, or MX). The formal theorems are proved in [Weinhold and Andersen \(2026\)](#); this document walks readers through the intuition behind why they hold. A single economic mechanism — *self-hedging* — organises the result: the rival's myopic removal of the most expensive items absorbs the cost pressure that value-first's selection would otherwise create. The formal analysis reveals three specific structural features of this mechanism: a four-level decomposition, a cross-damage symmetry governing the baseline independence case (yielding the closed-form asymptotic gap $\mu_E \phi - S_G(\tau) \approx 0.216$), and a two-force balance — selection amplification against cross-damage amplification — that shapes how the gap responds to correlation (generating a hump-shaped profile that remains positive across the feasible range and peaks near $\rho \approx 0.7$; certified margin ≥ 0.214 at $\rho = 0.3$). We close with a discussion of the valuation heterogeneity conditions under which the knapsack reversal holds and the scope of what the theorems formally establish.

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1 The big picture

Consider an allocation problem with two features. Items carry not one value but two: an ecological value and an agricultural value, neither reducible to the other. And the allocation is not made by a single planner but by two agents with different valuations, competing turn by turn to purchase items from a common pool. A conservation agency (Green) cares about the ecological value of items it purchases. An agricultural buyer (Farmer) cares about their agricultural value. Every item Farmer buys is one Green cannot, and every dollar Green spends raises the stakes of its remaining purchases.

The strategic environment produced by this conjunction — two values per item, two agents with distinct valuations, rivalrous turn-by-turn allocation — has its own logic. It is not the logic of the classical knapsack problem (which considers a single agent sorting items by value per unit cost), nor of standard procurement auctions (which consider a single-value contest with incentive design). [Weinhold and Andersen \(2026\)](#) formalise this environment and prove that within it, a simple rule — value-first, which tells Green to buy the item with the highest ecological value it can currently afford — outperforms the textbook cost-effectiveness rule (ratio-greedy, which tells Green to buy the item with the highest ecology per agricultural dollar). We call this phenomenon the *knapsack reversal*: the optimal heuristic in the classical knapsack problem is inverted when the problem becomes rivalrous and two-valued.

Throughout the paper, Farmers are modelled as *profit-maximising but naïve* — they buy the agriculturally most valuable available plot at each opportunity, but they do not observe the ecological value of plots and do not treat the conservation agency as a strategic rival. The term “naïve” refers to this informational and strategic limitation, not to any departure from canonical profit maximisation.

The knapsack reversal has a single economic mechanism behind it, which we call *self-hedging*. Green’s value-first selection of ecologically valuable items tends, under positive correlation between ecological and agricultural values, to create cost exposure: high- E items are typically also expensive. But Farmer, removing the most expensive items from the market turn by turn, systematically absorbs that cost pressure before it reaches Green. The adversarial structure that looks like a threat is, mechanically, providing a hedge — and the hedge comes for free

from Farmer’s myopia, with no action required by Green to secure it. The name “self-hedging” records this: the cost exposure that value-first creates is absorbed by the very rival activity the exposure would otherwise imply. Self-hedging is the overall mechanism behind the headline result. In formally establishing that this is a structural feature of adversarial rivalry in two-value space, the inner machinery of the mechanism reveals further economic insights — a four-level decomposition of how the hedge operates, a cross-damage symmetry governing the baseline case, and a two-force balance between selection amplification and cross-damage amplification that shapes how the gap responds to correlation.

The rest of this document provides intuition for the proofs. Explaining the proofs requires explaining the mechanism, and the document’s four sections develop both together. §2 states self-hedging in its general form, as four interlocking levels that jointly drive the knapsack reversal. §3 introduces the analytical lens through which the asymptotic gap can be tractably decomposed — a four-component view (prefix wedge, Farmer-active window, Green-only window, MX threshold) that the paper’s proofs all work with — and briefly reviews the three successive proof architectures through which the baseline theorem was established. §4 develops the baseline theorem in full, where independence and identical marginals give the mechanism its cleanest expression: a hidden symmetry of the game dynamics reduces the asymptotic gap to a one-line algebraic identity. §5 turns to the empirically relevant case of positive correlation, where the symmetry breaks but the mechanism survives: two economic forces (selection amplification, cross-damage amplification) shape how the gap depends on correlation, generating a hump-shaped profile that remains strictly positive across the entire feasible range and peaks near $\rho \approx 0.7$. §§6–7 close with two scope questions — when does valuation heterogeneity suffice for the knapsack reversal to hold, and what do the formal theorems do and do not establish?

2 The mechanism: self-hedging

The central economic idea is that *the naïve structure of the Farmer’s strategy hedges Green against the costs of its own success*. Green’s value-first rule, targeting high- E items, tends under positive correlation to be targeting expensive items as well — which ordinarily would create cost pressure. But Farmer, removing the most expensive items from the market turn

by turn, systematically absorbs that cost pressure before it reaches Green. The adversarial structure that looks like a threat is, mechanically, providing insurance. This is the self-hedging mechanism.

Self-hedging has four distinct levels. Under positive correlation all four are active and jointly drive the knapsack reversal. At $\rho = 0$ they collapse into a single clean symmetry, which is why the baseline case admits the remarkably tidy identity of §4.

Level 1 — value self-hedging. When the Farmer removes the most expensive items, those items are (under positive correlation) also ecologically valuable. Value-first has already secured some of the ecological elite in its opening moves; ratio-greedy was relying on those items being available later at a good ratio. Farmer’s removals systematically destroy ratio-greedy’s later options while leaving value-first’s portfolio intact. The quality gap between the two opening portfolios — the *prefix wedge*, the ecological gap between the top- E and top- E/A selections through the first mN Green moves — is the first engine of the knapsack reversal. It is the selection effect at the heart of the early-game comparison (§3) and widens as correlation rises (§5.1).

Level 2 — cost self-hedging. When Farmer removes the most expensive item from any pool, Green’s future spending on that pool cannot increase. Either Farmer’s removal targets an item Green intended to buy (because it also had high E), in which case Green substitutes a cheaper alternative — cheaper, because the removed item was the most expensive item available; or Farmer’s removal targets an item Green did not want, in which case Green is unaffected. Farmer’s interference can never raise Green’s costs. This is a structural property of the profit-maximising but naïve Farmer specifically: because Farmer is blind to ecological value and to the rivalry, it always removes the single most agriculturally valuable remaining item, and that item is (weakly) at least as expensive as anything Green would substitute. A strategic Farmer — one who observed ecological values, recognised the rivalry, and optimised removals to maximise damage to Green rather than to maximise its own near-term capture — could in principle target items whose removal forces Green into expensive substitutes. The knapsack reversal therefore depends on the game’s specific modelling of developer behaviour, and this level is where that dependence lives.

Level 3 — absorbing value interference through the fluid limit. Farmer’s deletions

do shift Green’s ecological opportunities — an item Green would have picked next may be gone — but the fluid-limit trajectory accounts for this automatically. Because the proof tracks Green’s *actual* per-turn ecological value along a deterministic curve derived from the survivor-pool law, Farmer’s interference is already priced into the trajectory. There is no separate “damage correction” to bound. This level is the economic content of the fluid-limit architecture introduced as the second of the three baseline proofs (§3), and it is what carries directly to the correlated case (§5.4), where the ODE is closed numerically rather than symbolically.

Level 4 — budget self-hedging. Under budget parity, both agents have enough money to purchase nearly everything, and the stopping share for value-first is determined by budget accounting rather than by strategic cost management. How many extra turns Green gets after Farmer exhausts reduces to a counting argument rather than a fragile cost-path calculation. In the proof at $\rho = 0.3$, this level appears as a deterministic budget floor $\beta_0 > 0.01$ between what Green’s budget can afford and what the fluid-limit trajectory costs; the stopping-time bootstrap uses this floor to guarantee that Green remains affordable through a certified horizon of the game.

All four levels reinforce the same conclusion: the adversarial structure rewards directness (buying the best ecology) and punishes cleverness (optimising the ratio), because the rival’s behaviour is entangled with the focal agent’s cost structure in a way that the static knapsack problem ignores.

The baseline symmetry is self-hedging in its purest form. When $\rho = 0$ and marginals are identical, Levels 1 and 3 collapse together into a clean symmetry: value-first’s picks are orthogonal to Farmer’s cost ordering, so Farmer’s removals do not destroy value-first’s options preferentially, and there is no value interference to absorb. The four-level machinery is not needed — the game has a closed-form solution and the gap is an algebraic identity. §3 reviews the three proof architectures through which this identity was successively discovered; §4 then develops the identity in full, and §5 returns to the four levels once symmetry breaks.

3 Three proofs, one theorem

Before any formal theorem work began, extensive computational simulations had established two things. First, the knapsack reversal — the phenomenon described in §2, in which value-first outperforms ratio-greedy under adversarial procurement — was a robust empirical pattern across every setting tested. Second, specific benchmark values had been pinned down. The normalised asymptotic gap between ME and MX was close to 0.216 under independence and close to 0.29 at $\rho = 0.3$. Farmer exhausted its budget at around turn $0.305 N$ along the ME path and around $0.296 N$ along the MX path. Value-first stopped buying near turn $0.695 N$; ratio-greedy a little later, near $0.704 N$. The self-hedging mechanism of §2 was visible, on every computational run, as a specific numerical pattern.

Formal analysis therefore did not begin by asking whether the knapsack reversal was true. It began by asking how to prove, rigorously, an effect that had already been characterised numerically to several decimal places. The numbers came first; the question was how close formal analysis could get to them.

3.1 The component view

What made formal analysis tractable was a decomposition of the asymptotic gap into four pieces. During the first mN Green purchases (the *prefix*, with $m = 0.10$), value-first and ratio-greedy select different portfolios; the ecological gap between these portfolios is the *prefix wedge*. Between the prefix and Farmer exhaustion (the *Farmer-active window*), both agents shop from a shrinking survivor pool, and value-first accumulates value at a per-turn rate set by the current ecological frontier. After Farmer exhausts (the *Green-only window*), value-first shops alone against a frozen Farmer frontier until its budget is spent. And the cost-effectiveness rule MX has its own asymptotic ceiling — the threshold $\Theta := \mathbb{E}[E \cdot \mathbf{1}\{A \leq q_F\}] - (\text{prefix})$ — determined by the joint distribution and the Farmer-exhaustion share ϕ . The asymptotic ME – MX gap is the sum of the prefix wedge and the Farmer-active plus Green-only windows, measured against Θ .

The three proofs of the baseline theorem that follow all work with some version of this

decomposition. They differ in how sharply they bound each component.

3.2 Three proofs, rising margins

The baseline theorem was proved three successive times, with increasingly tight certified margins.

The first proof bounded each component using a *worst-case deletion* principle: among all possible histories of prior removals from the survivor pool, the proof assumed the one most damaging to value-first had occurred. This gave a rigorous but pessimistic bound in every window, and the pessimism compounded over the roughly $0.2N$ contested turns of the game. The certified margin was ≥ 0.046 — a little over a fifth of the simulated gap. The theorem was proved; most of the economic signal was spent on deletion accounting.

The second proof replaced the worst-case bounds on the Farmer-active and Green-only windows with a *fluid-limit* argument. As the number of plots N grows large, the noisy discrete trajectory of cutoffs and budgets in the game settles into a smooth deterministic curve — its *fluid limit* — which can be analysed with calculus rather than stochastic methods. The key move of the second proof was to track what value-first *actually gets* per turn along this smooth trajectory, rather than to bound what it might lose to adversarial deletions. This required an ODE (ordinary differential equation) for the state of the game — the Farmer frontier, the Green cutoff, and the two remaining budgets, evolving continuously in the large- N limit — and a time integral of the per-turn ecological rate along that trajectory. Under independence with identical marginals, the ODE has a closed-form solution, and the integral can be evaluated symbolically. The prefix wedge was left in its conservative lower-bound form. The certified margin rose to ≥ 0.168 — close to 80% of the simulated gap.

The third proof observed that the same fluid-limit dynamics have a hidden *symmetry* under independence with identical marginals. Farmer picks by agricultural value A ; value-first picks by ecological value E . Because the two coordinates are independent and identically distributed, Farmer’s pick is ecologically random from value-first’s perspective, and value-first’s pick is cost-random from Farmer’s. The consequence is that Farmer’s per-turn ecological capture equals value-first’s per-turn agricultural spend at every moment of the contested phase. Integrating,

the two cumulative quantities agree exactly, and the asymptotic gap collapses to an algebraic identity:

$$\mu_E \phi - S_G(\tau) = 1.493871 - 1.277736 = 0.216135.$$

This is the exact asymptotic limit, not a lower bound. The margin is the true gap.

Successive refinement of this kind is the ordinary rhythm of mathematical work, not an artefact of this project’s methodology. Early proofs establish that a result is true using general, conservative machinery; later proofs find more specialised routes that extract more of the economic signal. The $0.046 \rightarrow 0.168 \rightarrow 0.216$ trajectory suggests roughly where the signal was hiding: most of it is in how sharply one can track value-first’s actual per-turn dynamics (the $0.168 - 0.046 = 0.122$ jump), and the remainder is in the baseline’s specialised symmetric structure (the $0.216 - 0.168 = 0.048$ jump).

3.3 What extends to positive correlation, and what does not

§4 will develop the third of these proofs — the symmetry identity — in full. It is the paper’s primary baseline proof and gives the cleanest economic reading: ratio-greedy cedes a ϕ -fraction of the market worth $\mu_E \phi$ in ecology, while value-first pays $S_G(\tau)$ in agricultural cost to race Farmer for those same items; the gap is the difference.

§5 then turns to positive correlation. The symmetry identity does not extend: at $\rho > 0$, Farmer’s $\arg \max A$ pick is correlated with high- E items, so Farmer’s per-turn ecological capture exceeds value-first’s per-turn agricultural spend, and no closed-form reduction is available. What does extend is the architecture of the second proof — the fluid-limit ODE, the component decomposition, and the survivor-pool law that underlies them. §5 will show that when the symmetry breaks, these are the tools that do the work, and that closure at $\rho = 0.3$ comes from validated numerical integration of the ODE rather than from a closed-form identity.

4 The baseline theorem ($\rho = 0$): the symmetry identity

This section develops the third of the three baseline proofs introduced in §3 — the symmetry identity — in full. Under independence with identical marginals, the game has a hidden symmetry that collapses the entire value calculation to a one-line identity for the asymptotic gap:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E}[\text{PC}_N^{\text{ME}} - \text{PC}_N^{\text{MX}}] = \mu_E \phi - S_G(\tau) \approx 0.216135.$$

The rest of this section explains where the symmetry comes from, why the identity has a clean economic reading as “racing versus ceding,” and how the numerical closure works.

4.1 The symmetric game

When $\rho = 0$ and both coordinates are $U[0.1, 10]$, the two agents are shopping in *orthogonal* dimensions. Farmer ranks items by agricultural value A ; Green (under ME) ranks them by ecological value E . Under independence, knowing an item’s rank on one dimension tells you nothing about its rank on the other. Farmer’s max- A pick is, from Green’s perspective, ecologically random; Green’s max- E pick is, from Farmer’s perspective, cost-random. We call this *agricultural blindness*: value-first neither avoids nor prefers expensive items — it selects on E with no cost awareness at all.

The consequence is symmetric market depletion. Each turn, Farmer removes the top- A item from the survivor pool and Green removes the top- E item. Under independence, the survivor pool after any number of these alternating removals is characterised by two cutoffs — a Farmer frontier on the A -dimension and a Green cutoff on the E -dimension — which evolve independently. In the fluid-limit coordinates $u = 1 - \Phi(x)$ and $v = 1 - \Phi(c)$ measuring the tail mass above each cutoff, the two frontiers satisfy identical equations of motion starting from identical initial conditions. So $u(s) = v(s)$ at every share s , and both follow the closed form

$$u(s) = v(s) = 1 - \sqrt{1 - 2s}.$$

This is symmetry as a property of the game, not an assumption of the proof. It is what agricultural blindness buys you.

4.2 The symmetry identity: racing versus ceding

The symmetry has an immediate economic consequence. At any share s , the items remaining in the pool have A -values uniform on $[0.1, x(s)]$ and (independently) E -values uniform on $[0.1, c(s)]$. Farmer’s max- A pick has A -value near $x(s)$ and an E -value drawn uniformly from $[0.1, c(s)]$ — Farmer takes a generic item’s worth of ecological value. Green’s max- E pick has E -value near $c(s)$ and an A -value drawn uniformly from $[0.1, x(s)]$ — Green pays a generic item’s worth of agricultural cost. By the symmetry $x(s) = c(s)$, these two “generic” averages are equal:

$$\text{Farmer's per-turn } E = \frac{1}{2}(0.1 + c(s)) = \frac{1}{2}(0.1 + x(s)) = \text{Green's per-turn } A.$$

This is the *cross-damage symmetry* of the baseline game: the ecological value Farmer captures from ME’s contested pool equals the agricultural cost Green incurs from Farmer’s contested pool. Each agent damages the other on the dimension it doesn’t care about, and under $\rho = 0$ with identical marginals, they do so in exactly equal measure. The importance of this observation is not that it says something deep about competition in general — it depends on both independence and identical marginals in a way that does not extend beyond this specification — but that it gives a clean reference point against which the correlated case (§5) can be compared. Positive correlation breaks this symmetry, and everything more complicated in the $\rho = 0.3$ proof is machinery for handling the breakdown.

Integrating the per-turn equality over the active phase: Farmer’s cumulative ecological capture through Farmer’s exhaustion share τ equals Green’s cumulative agricultural spend $S_G(\tau)$.

Now combine this with budget accounting. After Farmer exhausts, Green shops alone and (in the fluid limit) buys every remaining item. Green’s total ecological haul is therefore

$$\text{PC}^{\text{ME}}/N \rightarrow \mu_E - S_G(\tau),$$

because Green gets everything except what Farmer took, and Farmer’s ecological take equals $S_G(\tau)$ by symmetry. Ratio-greedy, on the other hand, concedes every item with A -value above Farmer’s budget-determined threshold q_F . Under independence with identical marginals that is exactly a ϕ -fraction of the market, each with average ecological value μ_E . So MX’s cession

to Farmer is worth $\mu_E \phi$ in ecological terms:

$$\text{PC}^{\text{MX}}/N \rightarrow \mu_E(1 - \phi).$$

Subtracting gives the identity

$$\mu_E \phi - S_G(\tau).$$

The gap is the difference between what MX *cedes by design* — the expensive ϕ -fraction of the market, worth $\mu_E \phi$ in ecology under independence — and what ME *loses by racing* — the cumulative agricultural cost $S_G(\tau)$ it pays by competing head-to-head with Farmer. In one phrase: *ratio-greedy retreats; value-first races; the race is worth the cost.*

This is self-hedging in its cleanest form. Level 1 is absorbed into the cross-damage symmetry itself (Farmer can damage value-first no more than Green damages Farmer). Level 2 is trivially operative but not load-bearing (there is no cost pressure to hedge against under independence). Level 3 is absent (no value interference to absorb). Level 4 shows up in the $\kappa = 1 - \tau$ identity, which says Green buys everything Farmer didn't.

4.3 What the identity reveals

Two features of the baseline identity deserve emphasis.

First, the identity is *exact*, not a bound: the equality $\lim_{N \rightarrow \infty} N^{-1} \mathbb{E}[\text{PC}^{\text{ME}} - \text{PC}^{\text{MX}}] = \mu_E \phi - S_G(\tau)$ is not an inequality. The other two proof architectures reviewed in §3 produced lower bounds that were conservative relative to the true gap, because they had to bound what value-first might lose from Farmer's interference and then close the remainder with explicit calculation. The symmetry argument does not need any such bound: agricultural blindness makes Farmer's and Green's per-turn captures distributionally identical, and the identity falls out of bookkeeping.

Second, the identity tells us where the knapsack reversal's advantage comes from at baseline. Because cross-damage between the two agents is symmetric — Farmer captures exactly as much ecological value from Green's pool as Green pays in agricultural cost from Farmer's — the advantage of value-first over ratio-greedy cannot, under independence with identical marginals,

come from Farmer being more damaging to one rule than to the other. It must come from elsewhere: from the fact that ratio-greedy voluntarily *cedes* the upper- A fraction of the market by design. That ceded portion is worth $\mu_E \phi$ in ecology, and the race value-first pays to compete for the remaining pool is only $S_G(\tau)$. The gap is the difference.

A useful way to remember the baseline result: at $\rho = 0$, adversarial cross-damage is symmetric, so the knapsack reversal’s advantage comes entirely from ratio-greedy’s own choice to retreat. At $\rho > 0$, this symmetry breaks — Farmer’s deletions begin to damage the two rules differently — and the richer dynamics of §5 take over.

4.4 Numerical closure

The identity $\mu_E \phi - S_G(\tau)$ involves two constants.

ϕ , the MX Farmer-exhaustion share. Defined by $\int_{1-\phi}^1 Q_A(v) dv = \mu_A/2$, which under the baseline uniform law reduces to the quadratic $9.9 \phi^2 - 20 \phi + 5.05 = 0$. The root in $(0, 1/2)$ is $\phi = (10 - \sqrt{50.005})/9.9 \approx 0.295816 \dots$

τ , the ME Farmer-exhaustion share. Defined by $S_F(\tau) = \mu_A/2 = 2.525$, where $S_F(s) = 0.1 s + 3.3 (1 - (1 - 2s)^{3/2})$. Since S_F is strictly increasing on $[0, 1/2]$, bisection on the bracket $[0.304718, 0.304719]$ locates τ to arbitrary precision.

Evaluating $\mu_E \phi = 5.05 \times 0.295816 \dots = 1.493871 \dots$ and $S_G(\tau) \in [1.277734, 1.277738]$ with $S_G(s) = 0.1 s + 1.65 (1 - (1 - 2s)^{3/2})$ gives

$$\mu_E \phi - S_G(\tau) \in [0.216133, 0.216137] = 0.21613 \dots > 0.$$

An 80-digit certificate based on *interval arithmetic* (arithmetic carried out on intervals rather than single numbers, so rounding error is bounded rather than hidden) tightens this to $[0.216134, 0.216136]$. The certificate is archived with the paper’s replication materials.

The visual shape of the argument is shown in Figure 1. Value-first builds its lead rapidly in the contested phase (while Farmer is still shopping), and although ratio-greedy recovers partially in the post-Farmer phase and purchases for slightly longer, the early lead is never erased.

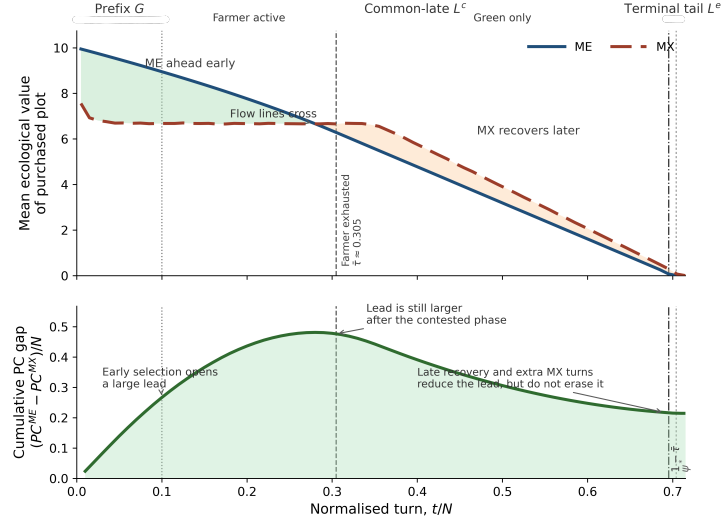


Figure 1: Visual anatomy of the baseline asymptotic dominance. Top panel: average ecological value of the plot purchased at each normalised turn under Max Environment and Max Efficiency in the baseline Budget World benchmark. The curves cross, so the theorem is not a pointwise-dominance result: Max Efficiency recovers later and purchases for slightly longer. Bottom panel: cumulative Purchased Conservation gap, $(PC^{ME} - PC^{MX})/N$. The contested phase (during which the Farmer is still shopping) creates a large advantage for Max Environment that the post-Farmer phase cannot undo. Vertical markers: prefix share $m = 0.10$, Farmer exhaustion $\bar{\tau} \approx 0.305$ in the fluid limit. Baseline $\rho = 0$, $N = 2000$, 3000 replications.

5 The correlated extension: when symmetry breaks

The baseline argument of §4 rests on a special feature of the independence case. When $\rho = 0$ and both coordinates have the same marginal distribution, Farmer’s $\arg \max A$ pick and Green’s $\arg \max E$ pick are, from each other’s perspective, random on the dimension the other cares about. The two agents shop in orthogonal dimensions: knowing an item’s rank on one dimension tells nothing about its rank on the other. Farmer’s removals of expensive items take ecologically generic items out of Green’s pool; Green’s removals of ecologically valuable items take agriculturally generic items out of Farmer’s. The damage each agent does to the other is symmetric in a precise sense, and that symmetry is what collapses the asymptotic gap to the closed-form identity $\mu_E \phi - S_G(\tau)$.

Under positive correlation between ecological and agricultural values, this orthogonality is lost. When high- E items tend to have high A as well, Farmer’s $\arg \max A$ pick is no longer ecologically generic — it is biased toward items Green would have valued. The agents are no longer shopping in separate aisles. Each of Farmer’s removals reduces the stock of high- E

items faster than under independence, so the pool from which Green draws becomes ecologically thinner over time at a rate that depends on Farmer’s activity rather than independently of it. The frontier dynamics that governed the baseline — two decoupled equations with a closed-form symmetric solution — become coupled equations that have no closed form.

This is not an obvious obstacle. A natural first instinct is that once the knapsack reversal is established for the independent case, the correlated extension should be a short step — correlation simply tilts the distribution, and the tilt should integrate out in a controlled way. In this case the instinct turns out to be wrong. The baseline’s phase-by-phase bookkeeping treats the Farmer-active and Green-only windows as near-separable because Farmer’s removals leave the pool ecologically generic; under correlation Farmer’s removals pull specifically on the high- E tail, and the two windows can no longer be analysed independently. The outside-prefix argument therefore has to be rebuilt around the fluid-limit architecture that §3 described as the second of the three baseline proofs.

What does carry over from baseline is the component decomposition of §3: prefix wedge, Farmer-active window, Green-only window, and MX threshold. Three of the four components remain static quantile integrals under the joint law — the same type of object as at baseline, merely evaluated under the Gaussian copula² rather than under independence. The fourth — the Farmer-active and Green-only windows, where the baseline symmetry was doing its work — is where the new architecture has to do it.

Two economic forces organise how the rebuilding plays out. *Selection amplification* strengthens the knapsack reversal: ratio-greedy’s corridor of cheap, high-ratio items becomes ecologically poorer as correlation rises, and the prefix wedge widens. *Cross-damage amplification* weakens it: Farmer’s correlated picks deplete the stock of high- E items faster than at baseline, and Green’s per-turn ecological capture drops accordingly. The two forces rise with ρ at different rates; their balance shapes the asymptotic gap across the feasible range. §§5.1–5.3 develop this picture economically, and §5.4 returns to the four components and walks through how each closes at $\rho = 0.3$.

²A Gaussian copula is a flexible way to specify a joint distribution from any two marginals (here both $U[0.1, 10]$) together with a correlation structure on an underlying Gaussian scale: latent variables (X, Y) are bivariate normal with correlation ρ , and $(A, E) = (0.1 + 9.9 \Phi(X), 0.1 + 9.9 \Phi(Y))$.

5.1 The selection force

Consider ratio-greedy’s corridor of high- E/A items — the portfolio MX selects in the prefix. At baseline, this corridor is ecologically respectable: high- E/A items are items with above-average E relative to their A , and since E and A are independent, these items’ E -values are distributed with only a mild upward tilt relative to the general population. As ρ rises, this changes. Positive correlation makes high E and high A go together: ecologically valuable items are typically also expensive. So “high E/A ” increasingly selects items where the denominator is anomalously low — cheap items, which under correlation are also ecologically mediocre. Ratio-greedy’s corridor becomes systematically ecologically impoverished.

Value-first is unaffected: it selects on E directly, regardless of A . The prefix wedge — the ecological gap between the top- E and top- E/A portfolios through the first $m = 0.10$ share of Green moves — therefore grows with ρ . At baseline the wedge is 0.219 (per normalised plot); at $\rho = 0.3$ it is 0.309. The wedge continues rising roughly linearly across the feasible range. This is Level 1 self-hedging (§2) strengthened under correlation: Farmer’s max- A aggression is no longer just removing expensive items — it is now removing items that value-first already secured, and the gap between what value-first captures early and what ratio-greedy captures early widens accordingly.

The selection force pushes the asymptotic gap *upward* as ρ rises.

5.2 The cross-damage force

Now consider what Farmer captures ecologically. At baseline, Farmer’s arg max A pick has an E -value distributed like a generic item in the survivor pool: under independence, knowing an item’s A -rank tells nothing about its E -value. As ρ rises, this changes. Positive correlation biases Farmer’s arg max A pick toward high E : the most expensive item in the pool tends to be (not always, but more often) one of the more ecologically valuable items. Farmer’s deletions therefore damage value-first more than at baseline — the items Farmer removes are increasingly items value-first wanted.

The rate at which this force grows is nonlinear in ρ . At low correlation, the bias is modest:

Farmer’s arg max A pick is still mostly an ecologically generic item. As ρ approaches 1, the bias becomes severe — at $\rho = 1$, Farmer’s arg max A pick *is* the arg max E item in the survivor pool, and Farmer captures the ecological maximum on every turn. The cross-damage force therefore accelerates toward the high-correlation limit.

The cross-damage force pushes the asymptotic gap *downward* as ρ rises.

5.3 The hump

The simulated gap $\Delta(\rho)/N = \mathbb{E}[\text{PC}_N^{\text{ME}} - \text{PC}_N^{\text{MX}}]/N$ is the net effect of these two forces. Since the selection force rises roughly linearly while the cross-damage force accelerates, the gap climbs through the intermediate range of ρ , peaks near $\rho \approx 0.7$ where the two forces balance at the margin, and declines thereafter as cross-damage overtakes. The gap remains strictly positive across the entire feasible range $[0, 0.995]$. Figure 2 shows the shape directly.

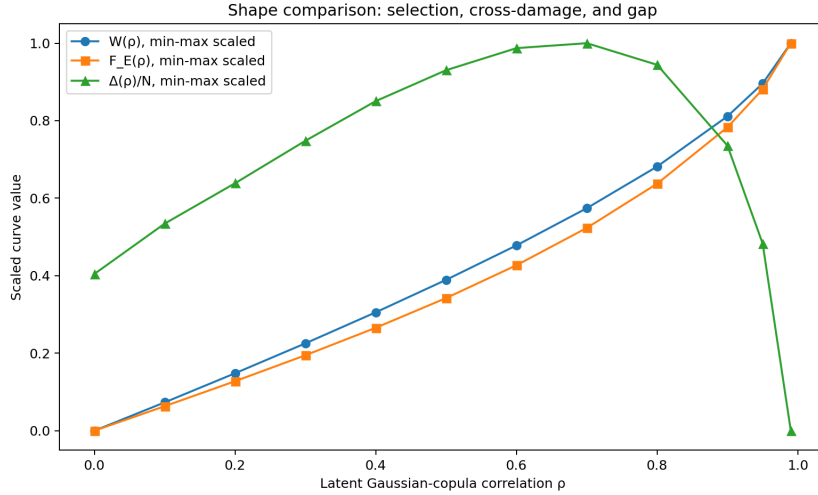


Figure 2: Two-force diagnostic for the hump-shaped ρ -gap. All three curves are min-max scaled to $[0, 1]$ for shape comparison. The selection proxy $W(\rho)$ (circles) rises approximately linearly in ρ , reflecting the growing ecological gap between the top- E and top- E/A portfolios. The cross-damage proxy $F_E(\rho)$ (squares) is gentle at low ρ and accelerates toward its $\rho \rightarrow 1$ limit, where Farmer captures the ecological maximum of the survivor pool on every turn. The simulated $\Delta(\rho)/N$ (triangles) rises while selection amplification dominates, peaks near $\rho \approx 0.7$, and declines as cross-damage overtakes. Simulation: $N = 2000$, 1000 replications per ρ , budget parity, Farmer moves first.

The policy reading is direct. The hump has two flanks: a rising flank from $\rho = 0$ through $\rho \approx 0.7$, and a falling flank from there toward $\rho = 1$. The empirically common case — moderately positive correlation between ecological and agricultural values — sits firmly in the rising flank.

The Bolivian landscape at $\rho \approx 0.29$ is in this flank. So is most of the conservation-economics evidence from other empirical settings, where production value and conservation value tend to be positively but imperfectly aligned. This is precisely the regime where cost-effectiveness mandates are most counterproductive: the knapsack reversal holds, and not barely — it is near its maximum.

A caveat on the decomposition. The selection proxy $W(\rho)$ and the cross-damage proxy $F_E(\rho)$ operate over different phases of the game and are not additively related to $\Delta(\rho)/N$. What the decomposition establishes is that the qualitative hump arises from the tension between two rising forces of opposite sign, not that $\Delta(\rho) = W(\rho) - F_E(\rho)$ pointwise. Figure 2 reports all three on a common min-max scaling for shape comparison only.

5.4 How the proof closes at $\rho = 0.3$

Theorem 4.2 of Weinhold and Andersen (2026) establishes the knapsack reversal at $\rho = 0.3$ formally, with a certified asymptotic margin of at least 0.214. The proof walks through the four components of the decomposition from §3.

The prefix wedge. As noted in §5.1, the wedge grows from 0.219 at baseline to 0.309 at $\rho = 0.3$. The computation is a pair of static quantile integrals under the Gaussian copula. The ME side uses a lower-envelope bound $\underline{M}_E(0.10) = 0.901$ carried over from the baseline; the MX side uses a threshold functional $M_X(0.10, 0.3) \leq 0.59222$ certified under the copula. The difference bounds the normalised prefix value gap below by 0.309 with probability approaching one.

The Farmer-active and Green-only windows. This is where the baseline symmetry no longer helps. Under correlation, value-first’s arg max E pick is biased toward high A — agricultural blindness is lost. A monotone-argmax lemma formalises this: conditional on the survivor pool, the probability that Green’s arg max E pick falls in the top- k block by A is at least k divided by the pool size. Green’s picks systematically bias toward agriculturally expensive items, accelerating the depletion of Farmer’s frontier, and Farmer’s picks symmetrically bias toward ecologically valuable items. The fluid-limit ODE that governs the game dynamics is therefore

coupled — the rate of change of each frontier depends on the other — and has no closed-form solution.

What remains is the ODE itself, which can be solved numerically. The state has four coordinates: the Farmer frontier on the A -dimension, the Green cutoff on the E -dimension, and the two remaining budgets. At every Green turn, the per-turn ecological value is determined by the current Green cutoff, and the cutoff evolves according to a fully explicit differential equation in those coordinates. Validated numerical integration (the interval-arithmetic approach of §4.4) establishes

$$\int_m^{\kappa_{\text{low}}} e(c(s)) ds \geq 2.5785$$

for the fluid-limit accumulation of ecological value from the end of the prefix through a certified affordable horizon $\kappa_{\text{low}} = 0.620816$. This is Level 3 self-hedging in its correlated form: the proof does not bound what Green might lose from Farmer’s interference, it tracks what Green actually accumulates along the ODE trajectory.

One technical foundation deserves explicit naming. The ODE rests on an *exact dynamic survivor law*: after any number of alternating max- A and max- E deletions, the pool of surviving items is exactly an iid sample from the doubly truncated Gaussian copula law, conditional only on the current two thresholds. This is what allows the state (x, c, b_F, b_G) to be Markov, which is what the density-dependent law of large numbers requires for the fluid-limit closure. Without the survivor law, the ODE would have no foundation; with it, the discrete game’s trajectory converges uniformly to the ODE’s solution as $N \rightarrow \infty$.

Affordability. The same ODE produces both the value side and the cost side. Integrating the per-turn agricultural spending rate along the ODE trajectory gives Green’s cumulative spend as a deterministic function of time. Through κ_{low} , this cumulative spend is at most 2.51477 — strictly less than Green’s starting budget $\mu_A/2 = 2.525$. So the ODE leaves a deterministic budget floor $\beta_0 \geq 0.01022$. A stopping-time bootstrap converts this deterministic floor into a finite- N affordability statement: as long as the discrete trajectory stays close to the ODE (which the density-dependent LLN delivers), Green remains affordable, and Green plays exact $\arg \max E$ throughout $[m, \kappa_{\text{low}}]$ with probability tending to one. This is Level 4 self-hedging in its correlated form — affordability is guaranteed by a deterministic margin between what the

budget can afford and what the game costs, not by a fragile cost-path calculation.

The MX threshold. The ratio-greedy limit reduces, as at baseline, to a static quantile integral: MX asymptotically purchases the set $\{A \leq q_F\}$, where $q_F = Q_A(1 - \phi)$ is the Farmer-exhaustion quantile. Under the Gaussian copula, the expectation $\mathbb{E}[E \cdot \mathbf{1}\{A \leq q_F\}]$ evaluates to a certified interval $[3.26511, 3.26513]$. To compare this against the ME outside-prefix integral on an apples-to-apples basis, we subtract the ME prefix lower envelope (0.901) that has already been counted toward value-first’s total. The resulting outside-prefix benchmark is $\Theta_{0.3} := \mathbb{E}[E \cdot \mathbf{1}\{A \leq q_F\}] - 0.901 \leq 2.36413$.

Closure. The Farmer-active plus Green-only ODE integral exceeds the outside-prefix benchmark by at least $2.5785 - 2.36413 = 0.21437$. This is the certified asymptotic margin at $\rho = 0.3$.

Why the margin is conservative. Simulation gives $\Delta(0.3)/N \approx 0.29$, so the certified 0.214 is roughly three-quarters of the true asymptotic gap. Two sources of conservatism remain in the formal argument. First, the prefix uses the ME lower envelope 0.901 rather than the exact prefix value. Second, the ODE integration stops at $\kappa_{\text{low}} = 0.620816$ rather than at full Green exhaustion; beyond κ_{low} the affordability bootstrap requires additional arguments not carried through here. Neither source affects the sign of the margin; both could in principle be tightened.

On the profit-maximising but naïve Farmer assumption. The cost-side argument above uses the ODE budget floor and the stopping-time bootstrap; it does not invoke a pathwise deletion-monotonicity lemma of the kind alluded to in §2 as Level 2 self-hedging. The observation that Farmer’s max- A deletions cannot increase Green’s future agricultural spending on any fixed pool — because the removed item is the most expensive available, and any substitute Green picks is weakly cheaper — is retained in the paper as *economic* interpretation of why the naïve-Farmer assumption matters. A strategic Farmer — one who observed ecological values, recognised the rivalry, and optimised removals to maximise damage to Green rather than to maximise its own A -capture — would require a different cost-side architecture. This is an open question that the current proof does not address.

6 When the knapsack reversal holds: valuation heterogeneity

The self-hedging mechanism requires one substantive ingredient: enough *valuation heterogeneity* — enough variation in the focal agent’s valuation of items across the pool — for the top-valuation portfolio and the top-ratio portfolio to actually differ. When valuations are nearly homogeneous — every item has roughly the same focal-agent value — value-first and ratio-greedy select similar portfolios and the knapsack reversal disappears. The prefix wedge shrinks, Level 1 stops doing work, and the architecture of the proof loses its driver.

The scalar diagnostic for this condition is the *Valuation Selection Premium* (VSP): the percentage by which the average focal-agent valuation of the top-valuation portfolio exceeds that of the top-ratio portfolio. In the conservation application studied here, the focal valuation is ecological value, and VSP specialises to the *ecological selection premium* (ESP):

$$\text{ESP} = \frac{\bar{E}_{\text{top-}E} - \bar{E}_{\text{top-}E/A}}{\mu_E}.$$

Compressed- E simulations indicate that the knapsack reversal holds empirically whenever ESP exceeds roughly 12.5% of mean E , while the current proof architecture requires approximately 15%. On the paper’s baseline grids, ESP is about 53%; on the Bolivian landscape, it exceeds 30%. Both are well above either threshold. ESP is straightforward to compute from any landscape dataset and offers practitioners a check on whether the mechanism is operative in their setting.

A structural observation sharpens the ESP story. Compressed- E simulations reveal that when ecological values are squeezed toward homogeneity, the stopping shares of the two rules hardly move — $K_N^{\text{ME}}/N \approx 0.695$ and $K_N^{\text{MX}}/N \approx 0.704$ across the full compression grid. What changes is not when the rules stop buying, but what they buy in their terminal turns: under compression, each of MX’s extra turns at the tail carries materially more ecological value than at baseline, because the compressed E -distribution concentrates value near the mean rather than in the upper tail. The proof architecture — the fluid-limit ODE, the affordability bootstrap, the component decomposition — is therefore structurally stable under ecological compression. What fails at low ESP is not the machinery but its driver: the prefix wedge shrinks toward zero, and Level 1 self-hedging loses its economic content.

The policy implication is direct: cost-effectiveness mandates are most counterproductive precisely where ecological and agricultural values are positively aligned and ecological heterogeneity is large. The Bolivian landscape is the empirically common case in both respects.

7 What the proofs show

The two theorems are proved for a specific data-generating process (uniform marginals on $[0.1, 10]$, Gaussian copula) at two correlation values. They do NOT prove:

- Universal dominance for all finite instances. Constructive counterexamples on very small grids (2×2 , 2×3) show that with tight budgets a single expensive asset can exhaust Green's budget, allowing ratio-greedy to outperform value-first. These pathological configurations are confined to grids where the budget concentration ratio $\kappa(N) = \max_i A_i / B_G$ is $O(1)$ rather than shrinking like $O_p(1/N)$. By 3×3 the pathology is already gone, and the Bolivian case study ($N = 390$, $\kappa \approx 0.01$) sits deep in the region where self-hedging dominates.
- Dominance at all correlation levels. Only $\rho = 0$ and $\rho = 0.3$ carry formal proofs. Simulations suggest the knapsack reversal holds across the entire feasible range $\rho \in [0, 0.995]$, with a hump-shaped profile peaking near $\rho \approx 0.7$.
- Dominance under strategic Farmers. The theorems use naïve max- A Farmers — profit-maximising on the single dimension they observe, but blind to the ecological dimension and to the strategic rivalry. Strategic Farmers — those observing ecological values, recognising the rivalry, and optimising over time rather than myopically — are studied only in simulations. The Level 2 economic content (Farmer's max- A deletions cannot raise Green's costs on any fixed pool) depends essentially on Farmer being naïve in this sense: a strategic Farmer could in principle target items whose removal forces Green into expensive substitutes. Whether the knapsack reversal survives under strategic Farmers is a question the current proof architecture does not address.
- Dominance under ecological homogeneity. When E -values are nearly homogeneous, ESP is small and the prefix wedge disappears. The theorems' conditions fail here, and simulation

evidence suggests the knapsack reversal itself fails.

8 Concluding reflection

The classical knapsack problem describes a single agent sorting items by value per unit cost from a fixed menu, and its canonical heuristic — ratio-greedy — is optimal under that framing. Adversarial procurement in two-value space differs on both counts: the focal buyer faces a rival whose purchases reshape the menu turn by turn, and each item’s cost to the buyer is the rival’s valuation of that item rather than a property intrinsic to the item. The paper formalises this environment and proves that within it, a simple rule that ignores cost entirely and selects directly on ecological value — value-first — asymptotically dominates the cost-effectiveness heuristic. This is the knapsack reversal, and the preceding sections have developed the intuition behind why it holds through progressively sharper views of the underlying mechanism.

The mechanism is self-hedging: the rival’s myopic removal of the most expensive items absorbs the cost pressure that value-first’s selection would otherwise create, with no action required by Green to secure the hedge. The formal analysis surfaced three specific structural features of this mechanism. A four-level decomposition — value, cost, value-interference, and budget self-hedging — describes how the hedge operates, and each level maps to a specific component of the formal proof. A cross-damage symmetry organises the baseline independence case: with both agents shopping in orthogonal dimensions, each damages the other by exactly the same amount on the dimension it does not care about, and the asymptotic gap reduces to the algebraic identity $\mu_E \phi - S_G(\tau) \approx 0.216$. A two-force balance organises the correlated case: selection amplification (ratio-greedy’s corridor becomes ecologically poorer as correlation rises) pushes the gap up, while cross-damage amplification (Farmer’s removals increasingly target items value-first wanted) pushes it down, and their net effect is a hump-shaped profile that remains strictly positive across the feasible range and peaks near $\rho \approx 0.7$.

The economic insights that these structural features surface are all about how rivalry, in a two-value setting, transforms the decision problem. Adversarial activity makes cost endogenous: under correlation, an item’s cheapness becomes an index of the rival’s disinterest, which is itself an index of lower intrinsic value. Ratio-greedy responds to this informational property

by retreating systematically toward items neither agent contests, and that retreat is precisely what the baseline identity captures. Value-first takes the opposite approach: it ignores cost entirely and lets the rival’s myopia absorb the resulting exposure. This is the core of self-hedging, and the inner machinery of the proofs — the four levels, the cross-damage symmetry, the two-force balance — describes where and how the mechanism operates. Level 2 tells us that the hedge depends specifically on profit-maximising but naïve developers; a strategic rival would change the analysis. Level 3 tells us that the damage a rival inflicts on Green’s future options is already priced into the value-first trajectory, so the knapsack reversal is robust to Farmer’s exact interference pattern. The cross-damage symmetry tells us that at the knife’s edge of independence and identical marginals, competitive interaction is an exact wash and the advantage comes entirely from ratio-greedy’s self-inflicted retreat. The two-force balance tells us that the empirically common regime — moderate positive correlation — is where the mechanism operates strongly.

Two broader observations close this document. First, adversarial rivalry in two-value space is a strategic environment with its own logic, governed by the interaction of rivalrous allocation and multi-dimensional valuation. The knapsack reversal is its headline result, but the richer finding is that a specific mechanism — self-hedging, with its four levels, its symmetry at baseline, and its hump under correlation — organises behaviour in this environment, with empirically testable conditions (valuation heterogeneity, positive correlation, naïve developers) that locate where the mechanism operates. Second, the formal analysis did more than establish that the knapsack reversal holds: in surfacing the structural features that make the mechanism work, it refined an initial intuition from computational simulation into something considerably richer. The proofs did not just verify a known result; they revealed what makes it hold, and that revelation is the economic content this document has attempted to distill.

References

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